Robustness of Information Circulation under Entropy Constraints

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In complex systems, cyclic causal structures are closely associated with recurrent and oscillatory dynamics, yet an integrated dynamical description remains lacking. From an information-theoretic perspective, we formulate such structures as information circulation and show that, in finite systems, the presence of a cycle is necessary for bounded, nontrivial dynamics. To characterize the underlying mechanism of this behavior, we analyze the merging of information flows at a single node. Near the entropy-capacity limit, competing inputs become effectively exclusive: increasing the pre-existing circulating flow suppresses additional incoming information. This establishes an intrinsic robustness mechanism under finite entropy constraints. The mechanism is validated in Gaussian multiple-access channels and the multiplex receivers, while it is only partially satisfied for AND gate interactions. Simulations of a binary stochastic network with sigmoid response further confirm the predicted suppression of external inputs by internal circulation. These results identify information circulation as a structural and dynamical basis for robustness in finite complex systems.

I. INTRODUCTION

Cyclic causal relationships are widely observed in complex systems, including gene regulatory networks[1–5], thalamocortical brain circuits[6], Arctic climate feedbacks[7], predator-prey ecosystems[8], and macro-financial systems[9]. These structures can be interpreted as feedback loops that support directed information circulation and often exhibit oscillatory or recurrent behavior. However, a unified and statistically grounded description of such cyclic causal relationships remains lacking, particularly for analyzing their dynamical properties.

To address this issue, we adopt an information-theoretic framework, which provides a model-independent and universally applicable description of correlations and causality in probabilistic systems. Within this framework, cyclic causal relationships can be formulated as cycles in an information dynamical network.

Information dynamics has been extensively developed across multiple disciplines, such as information thermodynamics[10–14], quantum thermodynamics[15–17], neuroscience[18,19], and bioinformatics[20]. These approaches describe entropy production in non-equilibrium systems and enable the identification of causal interactions in complex networks of neurons, genes, and proteins. In particular, mutual information and transfer entropy have been widely used to characterize correlations and directional information flow in many-body systems[12,21–25].

Building on these developments, Ref. 26 established a general and exact framework for information dynamics in which individual entropy, mutual information, transfer entropy, and reversed transfer entropy[14,27,28] are treated as the dynamical variables governed by a set of evolution equations with control variables. In that framework, the zero-control limit was analyzed, revealing that nonzero control variables induce curvature in the evolution of these dynamical variables and reflect the statistical structure of the underlying system.

Despite its generality, this framework does not explicitly address interactions between information flows, which are necessary to describe the dynamical behavior and responses of systems with cyclic structures.

In this paper, we extend the framework of Ref. 26 to describe cyclic causal relationships as cycles of transfer entropy, which encode directional causality[29]. Under locally deterministic dynamics and non-synergistic interactions, we demonstrate in Sec. III that a finite directed network of transfer entropy must contain a cycle in order to exhibit bounded and nontrivial dynamics. This result identifies information circulation as a necessary structural feature for such dynamics under the present assumptions.

To further characterize these dynamics, Sec. IV analyzes the behavior of a single cycle within the information dynamical framework. While the stationary state of an isolated cycle was previously studied in Ref. 26, we introduce interactions between the cycle and an external information source to investigate how such interactions modify the dynamics.

The analysis reveals that the behavior of the system depends critically on how control variables are assigned at points where multiple information flows merge. Accordingly, Sec. V examines the information dynamics at such merging points. Under a set of assumptions, we show that competing information flows become effectively exclusive when the individual entropy at the merging node is constrained, indicating a form of robustness of the information circulation.

II. INFORMATION DYNAMICAL EQUATIONS

A. Definitions

We summarize the information-theoretic quantities with increasing level of structure, from pairwise measures to directional and multi-variable extensions.

Definition 1 *The* ***Shannon entropy*** H_i *and* ***conditional entropy*** $H_j|_i$[30,31] *are defined as:*

$$\begin{aligned} H_i &:= -\sum_{\gamma_i \in \mathscr{M}} P(\Gamma_i = \gamma_i) ln\, P(\Gamma_i = \gamma_i) \\ H_j|_i &:= H_{ij} - H_i, \end{aligned} \tag{1}$$

where Γ_i *and* Γ_j *are discrete random variables taking values in the state space* $\mathscr{M}$.

Definition 2 *The* ***mutual information*** *between* Γ_i *and* Γ_j *is defined as*[31]*:*

$$I_{i:j} := H_i + H_j - H_{ij}, \tag{2}$$

and the ***conditional mutual information*** *is given by*

$$I_{i:j}|_k := H_i|_k + H_j|_k - H_{ij}|_k. \tag{3}$$

Definition 3 *The* ***transfer entropy***[21,32] *from j to i, which quantifies directional information flow, is defined as:*

$$T_{j\to i} := I_{i':j}|_i, \tag{4}$$

where i and i' *denote the random variables* $\Gamma_i(t_0)$ *and* $\Gamma_i(t_1)$*, respectively, with* $t_1 > t_0$.

Plus, all primed variables in this paper are evaluated one time step after the indicated base time.

Definition 4 *The* ***reversed transfer entropy***[14,27,28]*(also referred to as backward transfer entropy*[27]*) is defined as,*

$$rT_{j\to i} := I_{i:j'}|_{i'}, \tag{5}$$

which corresponds to the transfer entropy evaluated under time-reversed indexing. All quantities defined above are non-negative.

Definition 5 *The* ***multiple mutual information***[33–35] *captures multi-correlations, such as synergy and redundancy, and is defined as:*

$$I^n_{1:2:\ldots:n+1} := \sum_{i\in Y_1^{n+1}} H_i + \sum_{m=1}^{n} (-1)^m \sum_{(i_1,\ldots,i_{m+1})\in Y^{n+1}_{(m+1)}} H_{i_1,\ldots,i_{m+1}}, \tag{6}$$

where the index set $Y^n_l := \{(i_1,i_2,\ldots,i_l)|i_1,i_2,\ldots,i_l \in \{1,2,\ldots,n\}$ *and* $i_1 < i_2 < \cdots < i_l\}$ *for* $n \in \mathbb{N}$.

Definition 6 *The* ***multiple transfer entropy*** *and* ***multiple reversed transfer entropy*** *are defined as,*

$$\begin{aligned} T^n_{(i)_m\to(j)_l} &:= I^n_{j'_1:\cdots:j'_l:i_1:\cdots:i_m}|_{j_1,\ldots,j_l} \\ rT^n_{(i)_m\to(j)_l} &:= I^n_{j_1:\cdots:j_l:i'_1:\cdots:i'_m}|_{j'_1,\ldots,j'_l}, \end{aligned} \tag{7}$$

where the index set $(i)_m := \{i_1,i_2,\ldots,i_m\}$ *and* $(j)_l := \{j_1,j_2,\ldots,j_l\}$.

B. The General Framework of Information Dynamics

For a general multi-variable system with N components, we formulate the dynamical equations[26] governing the temporal evolution of the information-theoretic quantities defined in Sec. II A. In this framework, the entropy, mutual information, transfer entropy, and reversed transfer entropy are treated as state variables whose temporal changes are described by a set of coupled evolution equations.

Let ΔH_i, $\Delta I_{i:j}$, $\Delta T_{j\to i}$, and $\Delta rT_{j\to i}$ denote the temporal increments of the corresponding quantities over a discrete time step.

$$\begin{cases} \Delta H_i = \sum_{\eta\in K_i}[T_{\eta\to i} - rT_{\eta\to i}] - E_{K_i\to i} + \alpha_1^{K_i\to i} \\ \Delta I_{i:j} = [T_{j\to i} - rT_{j\to i}] + [T_{i\to j} - rT_{i\to j}] + \alpha_2^{ij} \\ \Delta T_{j\to i} = \sum_{\eta\in K_j-\{i\}}[T_{\eta\to j} - rT_{\eta\to j}] - [T_{j\to i} - rT_{j\to i}] - E_{K_j\to j} + \alpha_1^{K_j\to j} - \alpha_2^{ij} - \alpha_3^{j\to i} \\ \Delta rT_{j\to i} = \sum_{\eta\in K_j-\{i\}}[T_{\eta\to j} - rT_{\eta\to j}] - [T_{j\to i} - rT_{j\to i}] - E_{K_j\to j} + \alpha_1^{K_j\to j} - \alpha_2^{ij} + \alpha_4^{j} - \alpha_5^{ij} - \alpha_6^{j\to i}, \end{cases} \tag{8}$$

with,

$$\begin{aligned} &\alpha_1^{K_i\to i} := H_{i'}|_{i\ \zeta_1\ \dots\ \zeta_m} - H_i|_{i'\ \zeta_1'\ \dots\ \zeta_m'} \\ &\alpha_2^{ij} := I_{i':j'}|_{i\ j} - I_{i:j}|_{i'\ j'} \\ &\alpha_3^{j\to i} := \Delta H_j|_{i\ i'} \\ &\alpha_4^{j} := \Delta^2 H_j = \Delta(\Delta H_j) = H_{j''} - 2H_{j'} + H_j \\ &\alpha_5^{ij} := \Delta^2 I_{i:j} = I_{i'':j''} - 2I_{i':j'} + I_{i:j} \\ &\alpha_6^{j\to i} := \Delta H_{j'}|_{i\ i'}. \end{aligned} \tag{9}$$

The term $E_{K_i\to i}$ is responsible to the coupling between first-order(pairwise) correlations and high-order correlations:

$$\begin{cases} E_{K_i\to i} := \sum_{u=2}^{m}(-1)^u \sum_{(\eta)_u\subset K_i}[T^u_{(\eta)_u\to i} - rT^u_{(\eta)_u\to i}] \\ \Delta I^n_{\eta_1:\cdots:\eta_{n+1}} = \sum_{((i)_m,(j)_l)\in D_{K_\eta}}[T^n_{(i)_m\to(j)_l} - rT^n_{(i)_m\to(j)_l}] + \alpha_2^{S_\eta} \\ \Delta T^n_{(j)_m\to(i)_l} = (-1)^l \Delta I^n_{j_1:j_2:\cdots:j_m:i_1:i_2:\cdots:i_l} + \sum_{u=1}^{l}(-1)^{l-u}\sum_{(\eta)_u\subset(i)_l}(\Delta I^{n-u}_{j_1:\cdots:j_m:i_1:\cdots:\eta_1-1:\eta_1+1:\cdots:\eta_u-1:\eta_u+1:\cdots:i_l} \\ \quad + \sum_{s=1}^{u}(-1)^s \sum_{\{\zeta_1,\dots,\zeta_s\}\subset(\eta)_u}\alpha_3^{(j)_m\cup(i)_l-(\eta)_u\to(\zeta)_s}) \\ \Delta rT^n_{(j)_m\to(i)_l} = (-1)^l \Delta I^n_{j_1:j_2:\cdots:j_m:i_1:i_2:\cdots:i_l} + (-1)^l\alpha_5^{(j)_m\cup(i)_l} + \sum_{u=1}^{l}(-1)^{l-u}\sum_{(\eta)_u\subset(i)_l}(\Delta I^{n-u}_{j_1:\cdots:j_m:i_1:\cdots:\eta_1-1:\eta_1+1:\cdots:\eta_u-1:\eta_u+1:\cdots:i_l} \\ \quad + \alpha_4^{(j)_m\cup(i)_l-(\eta)_u} + \sum_{s=1}^{u}(-1)^s \sum_{\{\zeta_1,\dots,\zeta_s\}\subset(\eta)_u}\alpha_6^{(j)_m\cup(i)_l-(\eta)_u\to(\zeta)_s}), \end{cases} \tag{10}$$

with,

$$\begin{aligned} &\alpha_2^{S_\eta} := I^n_{\eta_1':\cdots:\eta_{n+1}'}|_{\eta_1\ \dots\ \eta_{n+1}} - I^n_{\eta_1:\cdots:\eta_{n+1}}|_{\eta_1'\ \dots\ \eta_{n+1}'} \\ &\alpha_3^{(r)_{n-u+1}\to(\zeta)_s} := \Delta I^{n-u}_{r_1:\cdots:r_{n-u+1}}|_{\zeta_1\ \zeta_1'\ \dots\ \zeta_s\ \zeta_s'} \\ &\alpha_4^{(r)_{n-u+1}} := \Delta^2 I^{n-u}_{r_1:\cdots:r_{n-u+1}} \\ &\alpha_5^{(r)_{n+1}} := \Delta^2 I^n_{r_1:r_2:\cdots:r_{n+1}} \\ &\alpha_6^{(r)_{n-u+1}\to(\zeta)_s} := \Delta I^{n-u}_{r_1':\cdots:r_{n-u+1}'}|_{\zeta_1\ \zeta_1'\ \dots\ \zeta_s\ \zeta_s'}. \end{aligned} \tag{11}$$

Each equation is exact[26] and consists of an intrinsic contribution, determined by the probabilistic structure of the system, and a control contribution, parameterized by coefficients α, which encode deviations from the zero-control limit. In the zero-control limit, these equations reduce to the intrinsic dynamics analyzed in Ref. 26. Furthermore, the control variables α encode how interaction structures and constraints modify the intrinsic information dynamics. Their values are determined by mechanisms such as blocked channels and finite-capacity effects at merging points.

Ref. 26 demonstrated that the non-trivial circulation near the zero-control limit arises from the restriction allowing the one-way interaction through one channel between particle A and B:

$$\begin{cases} [T_{B\to A}=0 \text{ and } rT_{A\to B}=0] \\ \text{or } [T_{A\to B}=0 \text{ and } rT_{B\to A}=0]. \end{cases} \text{for all time} \tag{12}$$

C. Structural Assumption : Deterministic Reversible Dynamics

The analysis of Sec. III assumes the deterministic reversible dynamics: $\exists\, U=(U_1,U_2,...,U_N)$ and U^{-1} such that,

$$\begin{aligned} \Gamma_i(t_2) &= U_i(\Gamma_1(t_1),\ldots,\Gamma_N(t_1),t_1) \\ \Gamma_i(t_1) &= U_i^{-1}(\Gamma_1(t_2),\ldots,\Gamma_N(t_2),t_2), \end{aligned} \tag{13}$$

where $i\in\{1,2,...,N\}$ and $t_2>t_1$. Then, the following uncertainty vanishes:

$$\begin{aligned} H_{i'}|_{1\ 2...N} &= 0 \\ H_i|_{1'\ 2'...N'} &= 0, \end{aligned} \tag{14}$$

where the primed variables are evaluated at time $t_2>t_1$. Hence, with $K_i:=\{1,2,...,N\}$, the control parameter $\alpha_1^{K_i\to i}$ defined in the equation (9) becomes zero in this case:

$$\alpha_1^{K_i\to i} := H_{i'}|_{1\ 2...N} - H_i|_{1'\ 2'...N'} = 0. \tag{15}$$

Furthermore, if one assumes the locally deterministic reversible dynamics, the set K_i can be effectively shrunk to the index set of the dynamical neighbors.

$$\begin{aligned} \Gamma_i' &= U_i(\Gamma_i,\Gamma_{k_1},\Gamma_{k_2},...,\Gamma_{k_m}) \\ \Gamma_i &= U_i(\Gamma_i',\Gamma_{k_1}',\Gamma_{k_2}',...,\Gamma_{k_m}'). \end{aligned} \tag{16}$$

With $K_i:=\{k_1,k_2,...,k_m\}\subset\{1,2,...,N\}$,

$$\alpha_1^{K_i\to i} := H_{i'}|_{ik_1k_2...k_m} - H_i|_{i'k_1'k_2'...k_m'} = 0. \tag{17}$$

D. Structural Assumption : Absence of Synergistic Interactions

Let the neighbor set of a particle v be $K=\{u_1,\ldots,u_m\}$, and define the notation:

$$\begin{aligned} T_{(u_1,...,u_m)\to v} &:= I_{u_1\ \ldots\ u_m:v'}|_v \\ rT_{(u_1,...,u_m)\to v} &:= I_{u_1'\ \ldots\ u_m':v}|_{v'}. \end{aligned} \tag{18}$$

This is distinct from the multiple transfer entropy and multiple reversed transfer entropy defined in Section II A.

Decomposing $T_{(u_1,...,u_m)\to v}$ and $rT_{(u_1,...,u_m)\to v}$ yields:

$$\begin{aligned} T_{(u_1,...,u_m)\to v} &= T_{u_j\to v} + I_{u_1\ \ldots\ u_{j-1}\ u_{j+1}\ \ldots\ u_m:v'}|_{v\ u_j} \\ rT_{(u_1,...,u_m)\to v} &= rT_{u_j\to v} + I_{u_1'\ \ldots\ u_{j-1}'\ u_{j+1}'\ \ldots\ u_m':v}|_{v'\ u_j'}, \end{aligned} \tag{19}$$

for $j\in\{1,\ldots,m\}$.

If $T_{(u_1,...,u_m)\to v}=0$ and $rT_{(u_1,...,u_m)\to v}=0$, then $T_{u_j\to v}=0$ and $rT_{u_j\to v}=0$ for all j, since the right hand terms are non-negative. Nevertheless, the converse is not generally true, as the counterexample is presented in Appendix. A. Restricting to non-synergistic interaction, the analysis of Sec. III assumes that the converse holds, i.e., if $T_{u_j\to v}=0$ and $rT_{u_j\to v}=0$ for all j, then $T_{(u_1,...,u_m)\to v}=0$ and $rT_{(u_1,...,u_m)\to v}=0$. This assumption excludes the pure synergistic effect, such as the XOR operation (Appendix. A), while it allows the cases of Gaussian MAC (Sec. V D 1) and additive sigmoid (Hill-type) response functions (Sec. VI).

III. GENERAL INFORMATION DYNAMICS OF A FINITE SYSTEM

We consider an N-particle system satisfying the blocking flows condition (Eq. (12)) and the assumption of deterministic reversible dynamics (Sec. II C). The system can be represented as a directed finite network, where edges correspond to pairwise transfer entropy (or reversed transfer entropy). To analyze the role of network structure, we distinguish between acyclic networks and networks containing at least one cycle. We recall the following definitions from graph theory[36,37].

Definition 7 *A* ***directed graph*** $(V,\overrightarrow{E})$ *is a graph* (V,E) *with an* ***orientation*** $\overrightarrow{E}$, *where* V *and* E *is a set of vertices and edges, and* $\overrightarrow{E} := \{(e,x,y)|e=xy\in E \text{ such that } x,y\in V \text{ with } x\neq y\}$. *Also, the reversed orientation* $\overleftarrow{E} := \{(e,y,x)|(e,x,y)\in\overrightarrow{E}\}$ *is defined. Then, the* ***in-degree*** *of a vertex* $v\in V$ *is the size of set* $\{(e,x,v)\in\overrightarrow{E}|e\in E \text{ and } x\in V\}$, *while the* ***out-degree*** *of a vertex* $v\in V$ *is the size of set* $\{(e,v,y)\in\overrightarrow{E}|e\in E \text{ and } y\in V\}$.

Definition 8 *A* ***source*** *in a directed graph* $(V,\overrightarrow{E})$ *is the vertex* $v\in V$ *with the zero in-degree, and a* ***sink*** *in a directed graph* $(V,\overrightarrow{E})$ *is the vertex* $v\in V$ *with the zero out-degree.*

Definition 9 *A **cycle** C in a directed graph $(V, \overrightarrow{E})$ is a subset of $\overrightarrow{E}$ such that $C := \{(e_1, v_1, v_2), (e_2, v_2, v_3), \ldots, (e_n, v_n, v_1)\}$ for vertices $v_1, \ldots, v_n \in V$ and $n \in \mathbb{N}$ with $n > 1$.*

Definition 10 *An **acyclic graph** is a directed graph without a cycle.*

The Proposition 1 and Proposition 2 show that acyclic information networks necessarily exhibit monotonic entropy evolution, preventing fully recurrent or stationary nontrivial dynamics under the present assumptions.

Proposition 1 *A finite acyclic graph must include at least one source and sink.*

(proof) From Proposition (3.19) in the chapter "Directed Acyclic Graphs and Topological Ordering" of Ref. 37, a finite acyclic graph $(V, \overrightarrow{E})$ must include at least one source. Since the directed graph with reversed orientation $(V, \overleftarrow{E})$ is also acyclic, it has a source w that is the sink in the original graph $(V, \overrightarrow{E})$. ■

While Proposition 1 is a classical result of graph theory, Proposition 2 focuses on the information dynamical aspect under the framework of II B, implying that acyclic information networks cannot sustain nontrivial dynamical behavior.

Proposition 2 *Under the standing assumptions of this section and the non-synergistic assumption of Sec. II D, if the directed network of pairwise transfer entropy is acyclic, then the individual entropy of a sink (or source) evolves monotonically in time.*

(proof) From Proposition 1, a directed graph of pairwise transfer entropy $(V, \overrightarrow{E})$ has a sink $v \in V$: $T_{u\to v} > 0$ and $T_{v\to u} = 0$ for all $u \in K$, where the set of v's neighbors $K := \{u \in V | (e, u, v) \in \overrightarrow{E}, e \in E\}$. Then, the dynamics of v's individual entropy (Eq. (8)) is,

$$\begin{aligned}\Delta H_v &= \sum_{u\in K} [T_{u\to v} - rT_{u\to v}] - E_{K\to v} + \alpha_1^{K\to v} \\ &= \sum_{u\in K} [T_{u\to v} - rT_{u\to v}] - E_{K\to v} \\ &= T_{(u_1,\ldots,u_m)\to v} - rT_{(u_1,\ldots,u_m)\to v},\end{aligned} \tag{20}$$

where $K = \{u_1, \ldots, u_m\}$, and here, $\alpha_1^{K\to v} = 0$ due to the assumption of deterministic and reversible dynamics.
Since $rT_{u_j\to v} = 0$ for all j from the blocking flows condition, the assumption of Section II D requires $rT_{(u_1,\ldots,u_m)\to v} = 0$. In addition, by definition, $T_{(u_1,\ldots,u_m)\to v} := I_{u_1\,\ldots\,u_m:v'}|_v \geqslant 0$. If $T_{(u_1,\ldots,u_m)\to v} = 0$, then $T_{u_j\to v} = 0$ for all j by the argument in Section II D, which contradicts the fact that v is the sink. Therefore,

$$\Delta H_v = T_{(u_1,\ldots,u_m)\to v} > 0. \tag{21}$$

For the case where the vertex v is a source, $\Delta H_v = -rT_{(u_1,\ldots,u_m)\to v} < 0$, indicating H_v decreases monotonically over time. ■

Such monotonic behavior corresponds to trivial dynamics, in the sense that the system cannot sustain oscillatory or stationary nontrivial states. If the entropy H_v is bounded, it must converge to a constant value. If unbounded, the dynamics cannot form an attractor, which is necessary for nontrivial behaviors, such as oscillations[26] or relaxation to a stable point. Therefore, the presence of a cycle is a necessary condition for nontrivial dynamics in finite information networks.

IV. INFORMATION DYNAMICS OF A FINITE SYSTEM WITH A CYCLE

In contrast to acyclic networks, finite systems containing a cycle can exhibit nontrivial dynamics. The information dynamics of such systems is analyzed within the framework of Sec. II, focusing on the behavior of a single cycle and its interaction with an external information source. Prior to that, the dynamics of an isolated cycle is considered.

A. The Stationary State of a Single Cycle

Let an N-particle system $\Gamma = \{\Gamma_0, \Gamma_1, \ldots, \Gamma_{N-1}\}$ form a one-dimensional system with periodic boundary condition. In the index notation, $\mod N$ is assumed in this section. By imposing the blocking flows condition and near the zero-control limit, Ref. 26 obtains,

$$\begin{aligned}&\Delta H_i = T_{i-1\to i} - rT_{i+1\to i} \\ &\Delta I_{i:i+1} = T_{i\to i+1} - rT_{i+1\to i} \\ &\Delta T_{i\to i+1} = T_{i-1\to i} - T_{i\to i+1} \\ &\Delta rT_{i\to i+1} = 0 \\ &\Delta T_{i\to i-1} = 0 \\ &\Delta rT_{i+1\to i} = rT_{i\to i-1} - rT_{i+1\to i}.\end{aligned} \tag{22}$$

These equations describe the circulation of transfer entropy and reversed transfer entropy along the cycle. A stationary state is achieved when the transfer entropy and reversed transfer entropy are uniformly distributed along the cycle, satisfying $T_{i-1\to i} = T_{i\to i+1}$, $rT_{i\to i-1} = rT_{i+1\to i}$, and $T_{i-1\to i} = rT_{i+1\to i}$. This corresponds to a constant circulation of information along the cycle. As shown in Ref. 38, small deviations from this stationary state remain bounded in the analyzed examples, suggesting the stability of the circulating information flow.

B. The Transient Behaviors of the Single Cycle and a Random Source

Consider the interaction between a single cycle and an external information source. Let an external particle *Ext* begin to interact with a particle p in the cycle at the time $t = t_1$ as shown in Fig. 1, while no interaction is present at earlier times. Therefore, the stationary state of the previous section is assumed for the time $t = t_0 < t_1$. Then, the resulting dynamics is described by Proposition 3.

Proposition 3

$$
\begin{aligned}
&\Delta H_i = T_{i-1\to i} - rT_{i+1\to i} \quad (i \neq p)\\
&\Delta H_p = T_{p-1\to p} + T_{Ext\to p} - rT_{p+1\to p} - E_{\{p-1,p+1,Ext\}\to p}\\
&\Delta MI_{ii+1} = T_{i\to i+1} - rT_{i+1\to i}\\
&\Delta MI_{pExt} = T_{Ext\to p} - rT_{p\to Ext}\\
\\
&\Delta T_{i\to i+1} = T_{i-1\to i} - T_{i\to i+1} \quad (i \neq p-1, p)\\
&\Delta rT_{i\to i+1} = 0\\
&\Delta T_{i\to i-1} = 0\\
&\Delta rT_{i\to i-1} = rT_{i-1\to i-2} - rT_{i\to i-1} \quad (i \neq p, p+1)\\
\\
&\Delta T_{p-1\to p} = T_{p-2\to p-1} - T_{p-1\to p} - \alpha_3^{p-1\to p}\\
&\Delta rT_{p\to p-1} = rT_{p-1\to p-2} - rT_{p\to p-1} - E'_{\{p-1,p+1,Ext\}\to p}\\
&\Delta T_{p\to p+1} = T_{p-1\to p} + T_{Ext\to p} - T_{p\to p+1} - E_{\{p-1,p+1,Ext\}\to p}\\
&\Delta rT_{p+1\to p} = \alpha_1^{\{p\}\to Ext} - \alpha_3^{Ext\to p} + rT_{p\to p-1} - rT_{p+1\to p}\\
&- E'_{\{p-1,p+1,Ext\}\to p} - \alpha_6^{p\to p-1}\\
\\
&\Delta T_{Ext\to p} = \alpha_1^{\{p\}\to Ext} - T_{Ext\to p} - \alpha_3^{Ext\to p}\\
&\Delta rT_{p\to Ext} = -\alpha_{3456}^{p\to Ext}
\end{aligned}
\tag{23}
$$

(proof) Appendix B.

These equations show that the external source modifies the information flow locally at particle p, where the incoming flow $T_{Ext\to p}$ merges with the circulating flow $T_{p-1\to p}$. The corresponding control variables determine how the newly activated interaction is incorporated into the pre-existing circulation.

Specifically, the terms, $\alpha_3^{p-1\to p}$, $\alpha_6^{p\to p-1}$, $\alpha_3^{Ext\to p}$, and $\alpha_6^{p\to Ext}$, are expected to become nonzero during the transient period after $t = t_1$, reflecting the local reorganization of information flow at the merging particle p. Since the transfer entropy $T_{Ext\to p} = 0$ for $t = t_0 < t_1$, the control variable $\alpha_3^{Ext\to p}$ determines the emergence of the nonzero transfer entropy $T_{Ext\to p}$ at later times. This interaction can therefore be interpreted as the merging of two information flows, $T_{Ext\to p}$ and $T_{p-1\to p}$, at the particle p. The control variables $\alpha_3^{Ext\to p}$ and $\alpha_3^{p-1\to p}$ govern the dynamics of this merging process.

V. THE MERGE OF TWO INFORMATION FLOWS

When a cycle of information flow interacts with an external information source, the manner in which two information flows merge at a single particle determines the resulting dynamics. The merging of two information flows is analyzed within the framework of Sec. II B.

A. Problem Setting

Two particles A and B interact with a particle p, while p also interacts with another particle q (Fig. 2). The states of these particles are denoted as $\Gamma_A, \Gamma_B, \Gamma_p$, and Γ_q, and they are random variables over an ensemble. At time $t = t_0$, the particles A, p, and q interact with each other, with the non-zero stationary transfer entropy $T_{A\to p}(t_0) = T_{p\to q}(t_0) = \mu_A \neq 0$ and reversed transfer entropy $rT_{p\to A}(t_0) = rT_{q\to p}(t_0) = \mu_A$, while the particle B is not interacting with the particle p, implying $T_{B\to p}(t_0) = 0$ and $rT_{p\to B}(t_0) = 0$. Then, the particles B and p start to interact at the time $t = t_1$. The quantities $T_{B\to p}(t_1)$ and $rT_{p\to B}(t_1)$ are determined from the dynamical equations (Eq. (8)).

$$
\begin{aligned}
&T_{B\to p}(t_1) = \Delta T_{B\to p}(t_0) = \sum_{\eta\in K_B-\{p\}} [T_{\eta\to B}(t_0) - rT_{\eta\to B}(t_0)]\\
&\quad - E_{K_B\to B}(t_0) + \alpha_1^{K_B\to B}(t_0) - \alpha_2^{Bp}(t_0) - \alpha_3^{B\to p}(t_0),\\
&rT_{p\to B}(t_1) = \Delta rT_{p\to B}(t_0)\\
&\quad = [T_{A\to p}(t_0) - rT_{q\to p}(t_0)] - E_{\{A,B,q\}\to p}(t_0)\\
&\quad + \alpha_1^{\{A,B,q\}\to p}(t_0) - \alpha_2^{Bp}(t_0) + \alpha_4^{p}(t_0) - \alpha_5^{Bp}(t_0) - \alpha_6^{p\to B}(t_0).
\end{aligned}
\tag{24}
$$

The conditions $T_{B\to p}(t_0) = rT_{p\to B}(t_0) = 0$ and the stationary conditions $\Delta I_{B:p}(t_0) = 0$ imply that $\alpha_2^{Bp}(t_0) = 0$. Plus, $E_{\{A,B,q\}\to p}(t_0) = E_{\{A,q\}\to p}(t_0) = 0$, as the particle B does not interact with particle p at time $t = t_0$.

$$
\begin{aligned}
&T_{B\to p}(t_1) = \sum_{\eta\in K_B-\{p\}} [T_{\eta\to B}(t_0) - rT_{\eta\to B}(t_0)] - E_{K_B\to B}(t_0)\\
&+ \alpha_1^{K_B\to B}(t_0) - \alpha_3^{B\to p}(t_0),\\
&rT_{p\to B}(t_1) = [T_{A\to p}(t_0) - rT_{q\to p}(t_0)] + \alpha_1^{\{A,B,q\}\to p}(t_0)\\
&+ \alpha_4^{p}(t_0) - \alpha_5^{Bp}(t_0) - \alpha_6^{p\to B}(t_0).
\end{aligned}
\tag{25}
$$

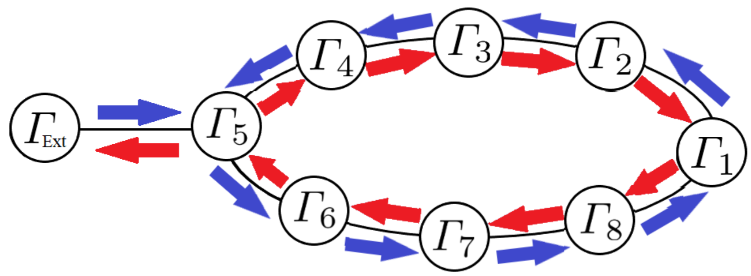

FIG. 1. Schematic illustrations of information dynamics on a cycle interacting with a random source (*Ext*). Blue arrows represent transfer entropy, and red arrows represent reversed transfer entropy.

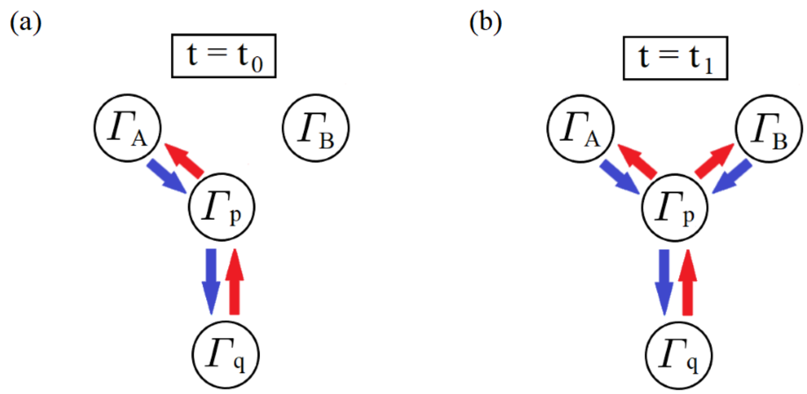

FIG. 2. Schematic illustration of the merging of two information flows at time **(a)** $t = t_0$ and **(b)** $t = t_1$. Blue arrows denote transfer entropy, and red arrows denote reversed transfer entropy.

B. The Zero-control Limit

The structural role of control variables α in the merge problem is recognized by analyzing the uncontrolled behavior of dynamical variables. In the zero-control limit with the blocking flow condition, all control variables vanish except those associated with blocked channels. Then, the equation (25) in the limit becomes,

$$T_{B\to p}(t_1) = \sum_{\eta\in K_B-\{p\}} [T_{\eta\to B}(t_0) - rT_{\eta\to B}(t_0)] - E_{K_B\to B}(t_0). \tag{26}$$

The transfer entropy $T_{B\to p}(t_1)$ is determined solely by incoming information flows and is not constrained by the entropy H_p. Since, $T_{A\to p}(t_1) = T_{A\to p}(t_0) = \mu_A$ and $rT_{q\to p}(t_1) = rT_{q\to p}(t_0) = \mu_A$ by the zero-control limit,

$$\Delta H_p(t_1) = [T_{A\to p}(t_1) + T_{B\to p}(t_1) - rT_{q\to p}(t_1)] = T_{B\to p}(t_1). \tag{27}$$

Consequently, the entropy H_p can increase without bound, indicating the absence of a stationary state with finite entropy. Nevertheless, it is not a general case. For instance, if the particle p has a binary state, then its entropy $H_p \leqslant ln\ 2$. In summary, the zero-control limit allows an unlimited increase of the particle p's entropy H_p, as the determination of transfer entropy $T_{B\to p}(t_1)$ is independent of the entropy H_p. Hence, the non-zero control variables α are required to introduce a finite maximum entropy of the particle p, the merging point.

C. Near the Zero-control Limit with a fixed maximum entropy

A finite upper bound on the entropy $H_p \leqslant ln\ Q$ imposes constraints on the merging dynamics. The regime $H_p \to ln\ Q$

is examined, under the following assumptions:

$$\begin{aligned}
&\text{Assumption 1}: T_{A\to p}(t_1) - T^2_{AB\to p}(t_1) = I_{A:p'}|_{pB}(t_1) = T_{A\to p}(t_0)\\
&\text{Assumption 2}: I_{q:p'}|_{pA}(t_0) = 0,\ I_{q:p'}|_{pAB}(t_1) = 0\\
&\text{Assumption 3}: \alpha_1^{\{A,q\}\to p}(t_0) = \alpha_1^{\{A,B,q\}\to p}(t_1)\\
&\text{Assumption 4}: \frac{\partial \alpha_1^{\{A,q\}\to p}(t_0)}{\partial T_{A\to p}(t_0)} > -1\\
&\text{Assumption 5}: \Gamma_q(t_1) = \Gamma_p(t_0)\\
&\text{Assumption 6}: \Delta I_{p':p}(t_0) + T_{B\to p}(t_1) \geqslant 0.
\end{aligned} \tag{28}$$

These assumptions isolate a minimal class of interactions where merging behavior can be analytically characterized.

From the equation (8),

$$\begin{aligned}
T_{B\to p}(t_1) =& \Delta H_p(t_1) + rT_{q\to p}(t_1) - T_{A\to p}(t_1)\\
&+ E_{\{A,B,q\}\to p}(t_1) - \alpha_1^{\{A,B,q\}\to p}(t_1).
\end{aligned} \tag{29}$$

The term $E_{\{A,B,q\}\to p}(t_1)$ can be reduced to the single multiple transfer entropy:

Proposition 4 *From Assumption 2, 5,*

$$E_{\{A,B,q\}\to p}(t_1) = T^2_{AB\to p}(t_1). \tag{30}$$

(proof) Appendix. C.

The Proposition 4 and Assumption 1 lead to:

$$T_{B\to p}(t_1) = \Delta H_p(t_1) + rT_{q\to p}(t_1) - T_{A\to p}(t_0) - \alpha_1^{\{A,B,q\}\to p}(t_1). \tag{31}$$

The limit $H_p \to ln\ Q$ is encoded by introducing $\varepsilon \geqslant 0$ such that,

$$\varepsilon := ln\ Q - \alpha_1^{\{A,q\}\to p}(t_0) - T_{A\to p}(t_0). \tag{32}$$

The limit $\varepsilon \to 0$ encodes that the sum of the transferred information from A and internal noise, $\alpha_1^{\{A,q\}\to p}$, reaches the maximum $ln\ Q$. Plus, the additional small δ is considered:

$$\delta := ln\ Q - \Delta H_p(t_1) - rT_{q\to p}(t_1), \tag{33}$$

indicating,

$$T_{B\to p}(t_1) = \varepsilon - \delta, \tag{34}$$

from Eq. (31) and Assumption 3.

Before discussing the relations of ε and δ, one can deduce the lemma.

Lemma 1 *From Assumption 1, 2, 3, 5, and Proposition 4,*

$$H_{p'}|_p(t_1) = H_{p'}|_p(t_0) + T_{B\to p}(t_1). \tag{35}$$

(proof) Appendix. D.

Then,

Proposition 5 *From Assumption 5, 6 and Lemma 1,*

$$\begin{aligned}
&(a)\ H_p(t_1) \geqslant ln\ Q - \varepsilon\\
&(b)\ 0 \leqslant \delta \leqslant \varepsilon\\
&(c)\ 0 \leqslant \Delta H_p(t_1) \leqslant \varepsilon.
\end{aligned} \tag{36}$$

(proof) Appendix. E.

From Proposition 5, one obtains,

$$0 \leqslant T_{B\to p}(t_1) \leqslant \varepsilon. \tag{37}$$

Hence, in the limit $\varepsilon \to 0$,

$$\begin{aligned}
&H_p(t_1) \to ln\ Q\\
&\Delta H_p(t_1) \to 0\\
&rT_{q\to p}(t_1) \to ln\ Q\\
&T_{B\to p}(t_1) \to 0
\end{aligned} \tag{38}$$

From this limit, $T_{B\to p}(t_1)$ can be expanded as:

$$T_{B\to p}(t_1)(\varepsilon) = \frac{\partial T_{B\to p}(t_1)}{\partial \varepsilon}(0)\varepsilon + O(\varepsilon^2), \tag{39}$$

assuming the differentiability of $T_{B\to p}(t_1)$ near $\varepsilon = 0$.
Given that Eq. (37),

$$\frac{\partial T_{B\to p}(t_1)}{\partial \varepsilon}(0) \geqslant 0, \tag{40}$$

indicating that $T_{B\to p}(t_1)$ is an increasing function of ε near the point $\varepsilon = 0$.

Since the Assumption 4 requires ε to decrease when $T_{A\to p}(t_0)$ increases, $T_{B\to p}(t_1)$ is regarded as a decreasing function of $T_{A\to p}(t_0)$ near the point $\varepsilon = 0$:

$$\frac{\partial T_{B\to p}(t_1)}{\partial T_{A\to p}(t_0)} = \frac{\frac{\partial T_{B\to p}(t_1)}{\partial \varepsilon}}{\frac{\partial T_{A\to p}(t_0)}{\partial \varepsilon}} \leqslant 0. \tag{41}$$

Equation (41) shows that, near the entropy-capacity limit of the merging node, an increase in the pre-existing circulating information flow suppresses the transfer entropy from the external source. Thus, in this limit, the internal circulation and the external input become effectively competitive. This provides an information-theoretic mechanism for robustness: a strong circulation protects the cycle by reducing the influence of perturbing inputs when the receiving node has limited entropy capacity. These arguments further indicate that the control variables α_3 and α_6, introduced previously, must satisfy conditions leading to Eq. (41) near the entropy-capacity limit.

In communication theory[39,40], this corresponds to the cross-talk effect in the two-senders-one-receiver multiple-access channels (MAC). The cross-talk[41–43] in MAC has been referred to as the difficulty of simultaneous communication with two senders, when the receiver's capacity for information processing is limited. Hence, the result of this section, Eq. (41) implies that the stronger the communication between particles p and $p-1$ is, the more difficult it becomes for particle Ext to communicate with particle p, if the individual entropy of particle p is near the upper limit. This cross-talk effect can be resolved when considering a multiplex reaction mechanism as in Sec. V D 2.

D. Examples of merging information flows

The validity of Eq. (41) and Assumption 1 $\sim$ 6 (Eq. (28)) are examined for specific dynamical systems.

1. Gaussian MAC

Let the dynamics at time $t<t_1$ be,

$$\begin{aligned}
&\Gamma_A(t) \sim N(0,\sigma_A)\\
&\Gamma_B(t) \sim N(0,\sigma_B)\\
&\Gamma_p(t) = \Gamma_A(t-1)\\
&\Gamma_q(t) = \Gamma_p(t-1),
\end{aligned} \tag{42}$$

where $N(\mu,\sigma)$ is the Gaussian distribution with the mean μ and variance σ^2, and σ_A and σ_B are positive.

At time $t \geqslant t_1$,

$$\begin{aligned}
&\Gamma_A(t) \sim N(0,\sigma_A)\\
&\Gamma_B(t) \sim N(0,\sigma_B)\\
&\Gamma_p(t) = \Gamma_A(t-1)+\Gamma_B(t-1)\\
&\Gamma_q(t) = \Gamma_p(t-1),
\end{aligned} \tag{43}$$

Since the distribution of $\Gamma_p(t)$ is the convolution of distributions of $\Gamma_A(t-1)$ and $\Gamma_B(t-1)$, one obtains $\Gamma_p(t) \sim N(0,\sqrt{\sigma_A^2+\sigma_B^2})$. Therefore,

$$\begin{aligned}
&T_{A\to p}(t_0) = ln(\sigma_A)+\frac{1}{2}ln(2\pi e)\\
&T_{B\to p}(t_1) = \frac{1}{2}ln\frac{\sigma_A^2+\sigma_B^2}{\sigma_A^2},
\end{aligned} \tag{44}$$

where the time $t_0<t_1$ and e is the natural constant.

This yields,

$$\frac{dT_{B\to p}(t_1)}{dT_{A\to p}(t_0)} = \frac{\frac{dT_{B\to p}(t_1)}{d\sigma_A}}{\frac{dT_{A\to p}(t_0)}{d\sigma_A}} = -\frac{\sigma_B^2}{\sigma_A^2+\sigma_B^2} < 0. \tag{45}$$

In this dynamics, the equation (41) is valid for the overall range. Also, it can be shown that the assumptions of Eq. (28) are valid.

2. A Multiplex Receiver

On the other hand, most artificial networks, such as multi source networks in communication theory, are designed to prevent the cross-talk effect for a credible communication. This is accomplished by modeling the multiplex receiver to process input signals independently. Therefore, the multiplex receiver is attributed two independent state space on the particle p: $\Gamma_p = (\Gamma_{p_1},\Gamma_{p_2})$. Then, the dynamics is set as,

$$\left.\begin{aligned}
&\Gamma_A(t) \sim Uniform(N_{Q_A})\\
&\Gamma_B(t) \sim Uniform(N_{Q_B})\\
&\Gamma_{p_1}(t) = \Gamma_A(t-1)\\
&\Gamma_{p_2}(t) = \Gamma_{p_2}(t-1)\\
&\Gamma_{q_1}(t) = \Gamma_{p_1}(t-1)\\
&\Gamma_{q_2}(t) = \Gamma_{p_2}(t-1),
\end{aligned}\right. \quad \text{for } \ t<t_1 \tag{46}$$

where $N_Q := \{1,2,\ldots,Q\}$ and Q_A, $Q_B \in \mathbb{N}$, and $Uniform(N_Q)$ is the uniform distribution for the set N_Q.

$$\left.\begin{aligned}
&\Gamma_A(t) \sim Uniform(N_{Q_A})\\
&\Gamma_B(t) \sim Uniform(N_{Q_B})\\
&\Gamma_{p_1}(t) = \Gamma_A(t-1)\\
&\Gamma_{p_2}(t) = \Gamma_B(t-1)\\
&\Gamma_{q_1}(t) = \Gamma_{p_1}(t-1)\\
&\Gamma_{q_2}(t) = \Gamma_{p_2}(t-1).
\end{aligned}\right. \quad \text{for } \ t \geqslant t_1 \tag{47}$$

As a result,

$$\begin{aligned}
&T_{A\to p}(t_0) = ln(Q_A)\\
&T_{B\to p}(t_1) = ln(Q_B),
\end{aligned} \tag{48}$$

and,

$$\frac{dT_{B\to p}(t_1)}{dT_{A\to p}(t_0)} = 0 \tag{49}$$

This represents a limiting case in which the cross-talk effect is eliminated, satisfying all assumptions of Eq. (28).

3. AND Gate

Unlike the previous examples, the merge with AND gate violates some of the assumptions and does not satisfy Eq. (41) in general.

$$\left.\begin{aligned}
&\Gamma_A(t) \sim B(\frac{1}{2}+\rho_A)\\
&\Gamma_B(t) \sim B(\frac{1}{2}+\rho_B)\\
&\Gamma_p(t) = \Gamma_A(t-1)\\
&\Gamma_q(t) = \Gamma_p(t-1),
\end{aligned}\right. \quad \text{for } \ t<t_1 \tag{50}$$

where $B(\rho)$ is the binomial distribution with probability ρ and $-\frac{1}{2} \leqslant \rho_A, \rho_B \leqslant \frac{1}{2}$.

$$\begin{aligned} &\Gamma_A(t) \sim B(\frac{1}{2}+\rho_A) \\ &\Gamma_B(t) \sim B(\frac{1}{2}+\rho_B) \qquad \text{for } t \geqslant t_1 \\ &\Gamma_p(t) = AND(\Gamma_A(t-1), \Gamma_B(t-1)) \\ &\Gamma_q(t) = \Gamma_p(t-1). \end{aligned} \tag{51}$$

Here,

$$\begin{aligned} &T_{A\to p}(t_0) = -(\frac{1}{2}+\rho_A)ln(\frac{1}{2}+\rho_A) - (\frac{1}{2}-\rho_A)ln(\frac{1}{2}-\rho_A) \\ &T_{B\to p}(t_1) \\ &= -(\frac{1}{2}+\rho_A)(\frac{1}{2}+\rho_B)ln(\frac{1}{2}+\rho_B) + (\frac{1}{2}-\rho_A)(\frac{1}{2}+\rho_B)ln(\frac{1}{2}-\rho_A) \\ &-(1-(\frac{1}{2}+\rho_A)(\frac{1}{2}+\rho_B))ln(1-(\frac{1}{2}+\rho_A)(\frac{1}{2}+\rho_B)). \end{aligned} \tag{52}$$

While,

$$\frac{\partial T_{B\to p}(t_1)}{\partial \rho_A} = (\frac{1}{2}+\rho_B)ln\frac{1-(\frac{1}{2}+\rho_A)(\frac{1}{2}+\rho_B)}{(\frac{1}{2}-\rho_A)(\frac{1}{2}+\rho_B)} \geqslant 0, \tag{53}$$

$$\frac{\partial T_{A\to p}(t_0)}{\partial \rho_A} = ln\frac{\frac{1}{2}-\rho_A}{\frac{1}{2}+\rho_A} \begin{cases} <0 & (\rho_A > 0) \\ >0 & (\rho_A < 0) \\ =0 & (\rho_A = 0) \end{cases} \tag{54}$$

Therefore,

$$\frac{dT_{B\to p}(t_1)}{dT_{A\to p}(t_0)} \begin{cases} \leqslant 0 & (\rho_A > 0) \\ \geqslant 0 & (\rho_A < 0) \end{cases} \tag{55}$$

The system satisfies the Assumption 2, 3, 5 clearly, however, Assumption 1 is invalid except the trivial case where $\rho_B = \frac{1}{2}$, and the validity of Assumption 4 depends on the value of ρ_A and ρ_B.

VI. EXAMPLE

To illustrate the mechanism derived in Sec. V, we consider a stochastic network with discrete state space. The merging node p represents a regulatory unit (e.g. a gene) that integrates inputs from A and B. Each node has a binary state, $\Gamma_i \in \{0,1\}$, and the update rule is given by a sigmoid response[44] to additive inputs,

$$P(\Gamma_p(t+1)=1) = \sigma(\beta_A\Gamma_A(t) + \beta_B\Gamma_B(t) + \kappa), \tag{56}$$

where $\sigma(x) = 1/(1+e^{-x})$ and $\kappa > 0$.

The update rule follows a standard logistic activation form, widely used in stochastic neural dynamics[45,46] and thermodynamic models of gene regulation[47,48], where the probability of activation depends on the combined influence of multiple inputs. In regimes where regulatory interactions are non-cooperative, thermodynamic models yield additive contributions from individual inputs, leading to sigmoidal activation functions of Hill type[47,48]. In such models, the additive structure arises from independent contributions of inputs, which combine linearly in the effective activation argument.

The remaining nodes in the cycle are modeled as delay-copy units, $\Gamma_{i+1}(t+1) = \Gamma_i(t)$, ensuring consistency with Assumption 5 in Sec. V. Within this construction, the cycle represents a persistent internal information circulation, while the node p acts as a finite-capacity processor where internal and external information flows compete.

To evaluate the resulting dynamics, we simulate an ensemble of independent realizations and estimate the relevant information-theoretic quantities using a simple binning method. The binary network is constructed with the cycle $p \to A1 \to \cdots \to A4 \to p$ and the external source Ext, and the particle p and Ext begin to interact after time $t = t_1$.

Simulation results (Fig. 3)[38] show that increasing the internal circulating flow $T_{A4\to p}(t_0)$ suppresses the external input $T_{Ext\to p}(t_1)$, quantitatively confirming the predicted relation (Eq. (41)).

VII. CONCLUSION

Various complex systems exhibit causal cycles that form feedback structures. Within the information-dynamical framework, these cycles are described as the circulation of transfer entropy and reversed transfer entropy. Under the present assumptions, this circulation constitutes a necessary structural condition for nontrivial dynamics in finite systems, including relaxation to stationary states and oscillatory behavior.

The stationary state of a single cycle, previously identified in Ref. 26, is extended here by incorporating interactions with an external information source. The resulting dynamics depends critically on how multiple information flows merge at a single particle. Under the assumptions of Eq. (28), the analysis shows that, near the limit of maximum entropy, strong circulating information flow suppresses the influence of external inputs.

This suppression mechanism corresponds to a cross-talk effect, in which limited entropy capacity constrains simultaneous information transfer, acting as a bottleneck. As a result, cycles with sufficiently strong internal circulation exhibit robustness against external perturbations.

These results indicate that finite complex networks require information circulation to sustain nontrivial dynamics, and that such circulation can provide robustness under entropy constraints. This perspective is consistent with observations

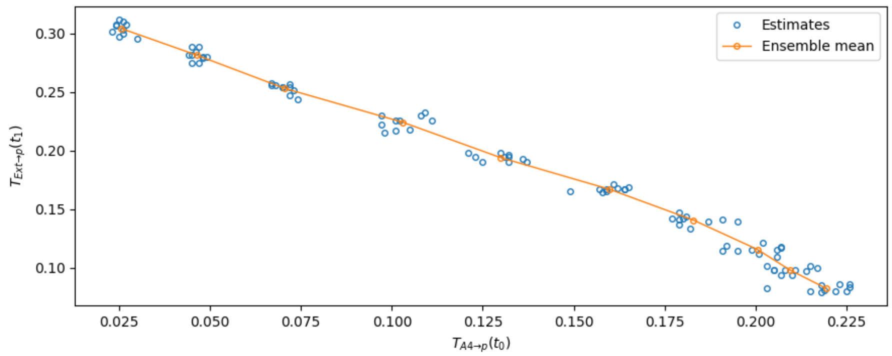

FIG. 3. Simulation results for competing information flows: $T_{A4\to p}(t_0)$ versus $T_{Ext\to p}(t_1)$. The cycle $p \to A1 \to \cdots \to A4 \to p$ is isolated at time $t = t_0$, and node p begins to interact with the external source (Ext) at $t = t_1$.

in gene regulatory networks, where cyclic structures are associated with robustness against external noise[4,5].

The analysis relies on several assumptions introduced in Sec. II and Eq. (28). While these assumptions enable tractable analysis of interacting information flows, their necessity and generality remain to be examined. Further work is required to relax these assumptions, explore a wider class of merging mechanisms, and investigate additional properties of information circulation, including its temporal structure and interactions between multiple cycles.

Furthermore, the present analysis assumes that the directed structure of transfer entropy interactions remains sufficiently stable over the analyzed time interval. Extending the framework to adaptive networks, in which the interaction topology and effective information flow structure evolve dynamically, remains an important direction for future work.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available in Zenodo at https://doi.org/10.5281/zenodo.20024405 (Version v0.1.3).

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Appendix A: The Counter Example of the Assumption of Section II D

Let the particles u_1, u_2, v with states Γ_{u_1}, Γ_{u_2}, and Γ_v in discrete time have the dynamics:

$$\Gamma_{u_1}(t_2), \Gamma_{u_2}(t_2) \sim B(\frac{1}{2}) \quad \Gamma_v(t_2) = \Gamma_{u_1}(t_1) \oplus \Gamma_{u_2}(t_1), \tag{A1}$$

where $t_2 > t_1$ and $B(p)$ is the binomial distribution with probability p, and the operation $\oplus$ represents the binary operation XOR.
Then, $H_{v'}|_v = H_{v'}|_{vu_1} = H_{v'}|_{vu_2} = ln\ 2$, resulting in $T_{u_1\to v} = T_{u_2\to v} = 0$. Nonetheless, $T_{(u_1,u_2)\to v} = H_{v'}|_v - H_{v'}|_{vu_1u_2} = ln\ 2$.

Appendix B: The Proof of Proposition 3

From the intrinsic method of Ref. 26, $\alpha_{3456}^{j\to i} := -\alpha_3^{j\to i} - \alpha_4^{j} + \alpha_5^{ij} + \alpha_6^{j\to i}$ can be expressed as:

$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{y} - \mathbf{A}^{-1}\mathbf{b}, \tag{B1}$$

where $\mathbf{y}, \mathbf{x}, \mathbf{b} \in \mathbb{R}^n$ and $\mathbf{A} \in \mathbb{R}^{n\times n}$ such that,

$$\begin{aligned}
(\mathbf{y})_l &= \alpha_3^{i\to j_l} - \alpha_6^{i\to j_l} - \Delta\alpha_2^{ij_l} \\
(\mathbf{x})_l &= \alpha_{3456}^{j_l\to i} \\
(\mathbf{b})_l &= \Delta E_{K_i\to i} - \Delta\alpha_1^{K_i\to i} \\
(\mathbf{A})_{lm} &= -1 + \delta_{lm},
\end{aligned} \tag{B2}$$

Here, the particle i's neighbor set is denoted as $K_i = \{j_1, j_2, ..., j_n\}$ with $n > 1$.
Therefore,
($i \neq p$ case)

$$\begin{aligned}
\alpha_{3456}^{i-1\to i} &= -\alpha_3^{i\to i+1} + \alpha_6^{i\to i+1} + \Delta E^{\{i-1,i+1\}\to i} \\
&\quad - \Delta\alpha_1^{i} + \Delta\alpha_2^{i\ i+1} \\
\alpha_{3456}^{i+1\to i} &= -\alpha_3^{i\to i-1} + \alpha_6^{i\to i-1} + \Delta E^{\{i-1,i+1\}\to i} \\
&\quad - \Delta\alpha_1^{i} + \Delta\alpha_2^{i\ i-1}.
\end{aligned} \tag{B3}$$

($i = p$ case)

$$\begin{pmatrix} \alpha_{3456}^{p-1\to p} \\ \alpha_{3456}^{p+1\to p} \\ \alpha_{3456}^{Ext\to p} \end{pmatrix} = \frac{1}{2}\begin{pmatrix} +1 & -1 & -1 \\ -1 & +1 & -1 \\ -1 & -1 & +1 \end{pmatrix}\mathbf{y} + \frac{1}{2}\mathbf{b}, \tag{B4}$$

where,

$$\begin{aligned}
\mathbf{y} &= \begin{pmatrix} \alpha_3^{p\to p-1} - \alpha_6^{p\to p-1} - \Delta\alpha_2^{p\ p-1} \\ \alpha_3^{p\to p+1} - \alpha_6^{p\to p+1} - \Delta\alpha_2^{p\ p+1} \\ \alpha_3^{p\to Ext} - \alpha_6^{p\to Ext} - \Delta\alpha_2^{p\ Ext} \end{pmatrix}, \\
\mathbf{b} &= [\Delta E_{\{p-1,p+1,Ext\}\to p} - \Delta\alpha_1^{\{p-1,p+1,Ext\}\to p}]\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.
\end{aligned} \tag{B5}$$

Substituting these relations into the dynamical equations (Eq. (8)) and imposing the blocking flows condition, $T_{i\to i-1}(t) = 0$, $rT_{i\to i+1}(t) = 0$, $T_{p\to Ext}(t) = 0$, $rT_{Ext\to p}(t) = 0$ for $\forall t, i$, yields,

$$\begin{aligned}
\Delta T_{i\to i+1} &= T_{i-1\to i} - T_{i\to i+1} - E_{\{i-1,i+1\}\to i} + \alpha_1^{\{i-1,i+1\}\to i} \\
&\quad - \alpha_2^{ii+1} - \alpha_3^{i\to i+1} \quad (i \neq p) \\
\Delta rT_{i\to i+1} &= 0 \\
\Delta T_{i\to i-1} &= 0 \\
\Delta rT_{i\to i-1} &= rT_{i-1\to i-2} - rT_{i\to i-1} - E'_{\{i-2,i\}\to i-1} + \alpha'^{\{i-2,i\}\to i-1}_1 \\
&\quad - \alpha'^{i-1i}_2 - \alpha_6^{i-1\to i-2} \quad (i \neq p+1) \\
\Delta T_{p\to p+1} &= T_{p-1\to p} + T_{Ext\to p} - T_{p\to p+1} - E_{\{p-1,p+1,Ext\}\to p} \\
&\quad + \alpha_1^{\{p-1,p+1,Ext\}\to p} - \alpha_2^{pp+1} - \alpha_3^{p\to p+1} \\
\Delta rT_{p+1\to p} &= \alpha_1^{\{p\}\to Ext} - \alpha_3^{Ext\to p} + rT_{p\to p-1} - rT_{p+1\to p} - \alpha_6^{p\to p-1} \\
&\quad - E'_{\{p-1,p+1,Ext\}\to p} + \alpha'^{\{p-1,p+1,Ext\}\to p}_1 - \alpha'^{pp-1}_2 \\
\Delta T_{Ext\to p} &= \alpha_1^{\{p\}\to Ext} - T_{Ext\to p} - \alpha_3^{Ext\to p} \\
\Delta rT_{p\to Ext} &= -\alpha_{3456}^{p\to Ext},
\end{aligned} \tag{B6}$$

where E' denotes for E evaluated at the following time step. As Ref. 26, it is required to analyze near the zero-control limit for understanding the impact of merge. Therefore,

$$\begin{aligned}
&\alpha_1^{\{i-1,i+1\}\to i} = \alpha_2^{ii+1} = 0 \quad (i \neq p) \\
&\alpha'^{\{i-1,i+1\}\to i}_1 = \alpha'^{ii+1}_2 = 0 \quad (i \neq p) \\
&\alpha_1^{\{p-1,p+1,Ext\}\to p} = \alpha'^{\{p-1,p+1,Ext\}\to p}_1 = \alpha_2^{ii+1} = \alpha'^{ii+1}_2 = 0 \\
&\alpha_3^{i\to i+1} = 0 \quad (i \neq p-1) \\
&\alpha_6^{i-1\to i-2} = 0 \quad (i \neq p+1),
\end{aligned} \tag{B7}$$

retaining only the control variables associated with the newly opened channel between Ext and p. In this limit, $\Delta E = 0$, as the high-order control variables also vanish. Therefore, the values of $E_{\{i-1,i+1\}\to i}$ and $E'_{\{i-2,i\}\to i-1}$ are assumed to be zero under sufficient initial conditions.

This results in the Eq. (23).
■

The equation (23) requires further specification. Firstly, $\alpha_{3456}^{p\to Ext}$ cannot be replaced by α_3, α_6 by the intrinsic method introduced in Ref. 26, since the particle Ext has only one neighbor. Therefore, $\alpha_{3456}^{p\to Ext}$ has to be determined from its definition, $\alpha_{3456}^{p\to Ext} = -\alpha_3^{p\to Ext} - \alpha_4^{p} + \alpha_5^{pExt} + \alpha_6^{p\to Ext}$. Secondly, defining the dynamic characteristics of the particle Ext can specify the terms, α_5^{pExt} and $\alpha_1^{\{p\}\to Ext}$. For instance, $\alpha_5^{pExt} = 0$ at all times and $\alpha_1^{\{p\}\to Ext} = 0$ at $t < t_1$ if the particle Ext is regarded as a source of white noise.

On the other hand, if $E_{\{p-1,p+1,Ext\}\to p} = 0$ permanently, there is no stationary state with non-zero $T_{Ext\to p}, rT_{p\to Ext}$. This shows that multivariate effects are essential for the existence of a stationary state with nonzero interaction with the external source.

Such transient effects are expected to appear in explicit dynamical realizations of the merging process. The transient behaviors after time $t = t_1$ and the way to approach another stationary state were demonstrated numerically in Ref. 38.

Appendix C: The Proof of Proposition 4

By the definition of E,

$$\begin{aligned}E_{\{A,B,q\}\to p} &= [T^2_{AB\to p} - rT^2_{AB\to p}] + [T^2_{Aq\to p} - rT^2_{Aq\to p}] \\ &\quad + [T^2_{Bq\to p} - rT^2_{Bq\to p}] - [T^3_{ABq\to p} - rT^3_{ABq\to p}].\end{aligned} \tag{C1}$$

From Assumption 5, one obtains,

$$\begin{aligned}rT^2_{Aq\to p} &= I^2_{A':q':p}|_{p'} = I_{A':q'}|_{p'} - I_{A':q'}|_{pp'} \\ &= I_{A':p}|_{p'} \quad (\because \text{Assumption 5: } q' = p) \\ &= rT_{A\to p} = 0 \quad (\because \text{blocking flows condition}),\end{aligned} \tag{C2}$$

and $rT^2_{Bq\to p} = 0$ in the similar manner. In addition,

$$\begin{aligned}rT^3_{ABq\to p} &= I^3_{A':B':q':p}|_{p'} = I^2_{A':B':p}|_{p'} - I^2_{A':B':p}|_{p'q'} \\ &= I^2_{A':B':p}|_{p'} \quad (\because \text{Assumption 5: } q' = p) \\ &= rT^2_{AB\to p}.\end{aligned} \tag{C3}$$

The $T^3_{ABq\to p}$ is estimated with the Assumption 2:

$$\begin{aligned}T^3_{ABq\to p} &= I^3_{A:B:q:p'}|_p = I^2_{A:q:p'}|_p - I^2_{A:q:p'}|_{pB} \\ &= T^2_{Aq\to p} - I_{q:p'}|_{pB} + I_{q:p'}|_{pAB} \\ &= T^2_{Aq\to p} + [I_{q:p'}|_p - I_{q:p'}|_{pB}] + I_{q:p'}|_{pAB} \\ &\quad (\because I_{q:p'}|_p = T_{q\to p} = 0) \\ &= T^2_{Aq\to p} + T^2_{Bq\to p} \\ &\quad (\because I_{q:p'}|_{pAB} = H_{p'}|_{pAB} - H_{p'}|_{pABq} = 0).\end{aligned} \tag{C4}$$

By summing up above results, the Eq. (C1) becomes,

$$E_{\{A,B,q\}\to p} = T^2_{AB\to p}. \tag{C5}$$

■

Appendix D: The Proof of Lemma 1

By the Assumption 2 and 5,

$$\begin{aligned}\alpha_1^{\{A,q\}\to p}(t_0) &= H_{p'}|_{pAq} - H_p|_{p'A'q'} = H_{p'}|_{pA}(t_0) \\ \alpha_1^{\{A,B,q\}\to p}(t_1) &= H_{p'}|_{pABq} - H_p|_{p'A'B'q'} = H_{p'}|_{pAB}(t_1).\end{aligned} \tag{D1}$$

Since,

$$\begin{aligned}H_{p'}|_p(t_0) &= T_{A\to p}(t_0) + H_{p'}|_{pA}(t_0) \\ &= T_{A\to p}(t_0) + \alpha_1^{\{A,q\}\to p}(t_0),\end{aligned} \tag{D2}$$

$$\begin{aligned}H_{p'}|_p(t_1) &= T_{B\to p}(t_1) + T_{A\to p}(t_1) - T^2_{AB\to p}(t_1) + H_{p'}|_{pAB}(t_1) \\ &= T_{B\to p}(t_1) + T_{A\to p}(t_0) + H_{p'}|_{pAB}(t_1)(\because Assumption 1) \\ &= T_{B\to p}(t_1) + T_{A\to p}(t_0) + H_{p'}|_{pA}(t_0)(\because Assumption 3) \\ &= T_{B\to p}(t_1) + H_{p'}|_p(t_0).\end{aligned} \tag{D3}$$

■

Appendix E: The Proof of Proposition 5

(a)
From Eq. (D2),

$$\begin{aligned}H_{p'}|_p(t_0) &= T_{A\to p}(t_0) + \alpha_1^{\{A,q\}\to p}(t_0) \\ &= ln\ Q - \varepsilon,\end{aligned} \tag{E1}$$

one obtains,

$$H_p(t_1) = H_{p'}(t_0) \geqslant H_{p'}|_p(t_0) = ln\ Q - \varepsilon. \tag{E2}$$

(b)
From Assumption 5,

$$rT_{q\to p}(t_1) = H_{q'}|_{p'} - H_{q'}|_{pp'} = H_p|_{p'} = -\Delta H_p + H_{p'}|_p. \tag{E3}$$

As a result,

$$H_{p'}|_p(t_1) = ln\ Q - \delta. \tag{E4}$$

Since $H_{p'}|_p(t_1) \leqslant H_{p'}(t_1) \leqslant ln\ Q$, one obtains $0 \leqslant \delta$.
Here, $\delta \leqslant \varepsilon$ is derived from the non-negativity of $T_{B\to p}(t_1)$.

(c)
From Assumption 6 and Lemma 1,

$$\Delta I_{p':p}(t_0) = \Delta H_{p'}(t_0) - \Delta H_{p'}|_p(t_0) = \Delta H_p(t_1) - T_{B\to p}(t_1) \geqslant -T_{B\to p}(t_1). \tag{E5}$$

Therefore,

$$\Delta H_p(t_1) \geqslant 0. \tag{E6}$$

Then,

$$ln\ Q \geqslant H_{p'}(t_1) \geqslant H_p(t_1) \geqslant ln\ Q - \varepsilon. \tag{E7}$$

Hence,

$$\Delta H_p(t_1) \leqslant ln\ Q - (ln\ Q - \varepsilon) = \varepsilon. \tag{E8}$$

■